DrozyAI Progress Report

Data & Alignment (Person 1)

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covering Person 1 and Person 2

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1 Summary

This report covers the completion of **Person 1** workstream: dataset acquisition, license verification, KSS label table construction, and the aligned per-second table that feeds the wearable model (Person 2) and the vision/fusion pipeline (Person 3).

Completed

Person 1 deliverable is complete.

- `data/processed/full_dataset.parquet` (84,000 rows × 16 columns)
- 35 sessions aligned to 1-second resolution
- Label distribution: 62.6% alert / 37.4% drowsy
- Loader function ready for import by other teammates
- Code pushed to GitHub

2 Completed Work

2.1 Dataset Acquisition and Structure

- Downloaded and inspected the **UL-DD dataset** (University of Louisiana Drowsiness Dataset).
- **19 subjects** (A–S), each with 1–2 sessions.
- **35 usable sessions** total:
  - Awake session (A): 19
  - Drowsy session (D): 16
- Each session is 40 minutes; **1,400 minutes** of raw data in total.
- Data stored locally under `data/raw/CSV_Files/CSV_Files/{A..S}/{A,D}/`.

2.2 License and Citation

- UL-DD is published in **Scientific Data** (Nature portfolio), vol. 13, article 506 (2026).
- Hosted on **Zenodo**, free for academic use.
- Citation: Bodaghi et al., *A multimodal drowsiness dataset using video, biometric, and behavioral data*, Sci Data 13, 506 (2026).

2.3 Label Table

- Each session has **10 KSS values**, one every 4 minutes, covering the full 40 minutes.
- **Binary threshold defined**: **KSS $\geq$ 7 = drowsy (1)**, KSS $<$ 7 = alert (0).
- Threshold documented in README.

2.4 Aligned Per-Second Table

All signals were resampled to **1-second steps**:

Signal	Source rate	Aggregation to 1 s
HR	1 Hz	Kept as-is
TEMP	4 Hz	Per-second mean

Signal	Source rate	Aggregation to 1 s
EDA	4 Hz	Per-second mean
ACC	32 Hz	Per-second mean and variance (per axis)
IBI	Irregular	Per-second mean and sample count
Labels	10/session	Expanded to per-second via 240-s intervals

Output files:

- `data/processed/{user}_{session}.parquet` (35 files, one per session)
- `data/processed/full_dataset.parquet` (**84,000 rows × 16 columns**)

2.5 Data-Quality Report

- **Label distribution**: alert (0) = 52,560 rows, drowsy (1) = 31,440 rows → **62.6%** / **37.4%**, balanced.
- **Missing values**:
  - Core signals (HR, TEMP, EDA, ACC): only 296 NaN (0.35%), caused by a few short sessions.
  - `ibi_mean`: 68% missing, which is **expected** (IBI is per-beat, not per-second). Person 2 will use `ibi_count` to gate the RMSSD computation.
- **Known issues**:
  - Subject J's awake session has an empty IBI file (0 bytes); the loader handles it gracefully.
  - 4 subjects (C, F, L, ...) have only the awake session.

2.6 Code and Loader

New files added to the repository:

File	Purpose
`src/data/loader.py`	Read raw CSVs, align to 1 s, expand KSS
`src/data/build_dataset.py`	Batch-process 35 sessions to Parquet
`src/main.py`	Runs the Person 1 pipeline

Loader usage:

```
from src.data.loader import load_session
df = load_session("A", "D",
                  Path("data/raw/CSV_Files/CSV_Files"))
```

2.7 Push to GitHub

- Repository: `git@github.com:fleurinejaegler/DrozyAI.git`
- Commit: `a5d88c3 Fix .gitignore anchors and add Person 1 pipeline code`
- **Data is not pushed** (18 GB, excluded via `.gitignore`).

3 Handoff

Handoff

To Person 2 (wearable model).

- `data/processed/full_dataset.parquet` (84,000 rows)
- Feature columns: `heart_rate`, `temp`, `eda`, `accel_mean_{x,y,z}`, `accel_var_{x,y,z}`, `ibi_mean`, `ibi_count`
- Label columns: `kss`, `drowsy` (KSS $\geq 7 = 1$)
- Group columns: `subject`, `session`, `second`
- Loader: `from src.data.loader import load_session`

Handoff

To Person 3 (vision / fusion). Each record carries `subject`, `session`, `second`, so vision features can be joined on these keys.

4 Open Items and Risks

Open Items / Risks

Item	Status
Data alignment	Done
Data-quality report	Done
README update (KSS threshold, dataset source)	Pending
Person 2 wearable model	Starting next
4 subjects missing drowsy session	Known; use available sessions
`ibi_mean` 68% missing	Known; handled via `ibi_count`

5 Key Links

- Dataset: Zenodo (see citation)
- Paper: Bodaghi et al., Sci Data 13, 506 (2026)
- GitHub: https://github.com/fleurinejaegler/DrozyAI

6 What Is on GitHub

The repository at https://github.com/fleurinejaegler/DrozyAI contains **code and documentation only**. All large datasets (18 GB) and the third-party reference code (95 MB) are intentionally excluded via `.gitignore`.

6.1 Repository Tree

The following tree shows every file currently tracked by Git:

```
DrozyAI/
 .gitignore
```

```
README.md
requirements.txt
docs/
notebooks/
src/
   __init__.py
   main.py
   data/
      __init__.py
      loader.py
      build_dataset.py
   capture/
      __init__.py
      webcam.py
      watch.py
   features/
      __init__.py
      watch_features.py
      video_features.py
      fusion.py
   models/
      __init__.py
      baseline.py
      gru.py
      calibration.py
   advisor/
      __init__.py
      nemotron_client.py
      fallback.py
   dashboard/
      app.py
   driver_app/
       app.py
tests/
```

6.2 File-by-File Description

Path	Purpose
`.gitignore`	Defines what Git ignores. Anchors `/data/` and `/reference/` to the repository root, so files under `src/data/` are still tracked.
`README.md`	Short project description. To be expanded with KSS threshold, dataset citation, and setup instructions.
`requirements.txt`	Python dependencies: `numpy`, `pandas`, `scipy`, `scikit-learn`, `torch`, `opencv-python`, `mediapipe`, `streamlit`, `plotly`, `bleak`, `python-dotenv`.
`docs/`	Project documentation (currently empty; reserved for the data-quality report and usage notes).

Path	Purpose
`notebooks/`	Exploratory Jupyter notebooks (currently empty).
`src/__init__.py`	Marks `src` as a Python package.
`src/main.py`	Top-level entry point. Currently invokes the Person 1 pipeline (`build_dataset`).
`src/data/loader.py`	Reads the raw UL-DD CSVs, resamples every signal to 1 Hz, and expands the 10 KSS values into a per-second label. *Deliverable of Person 1.*
`src/data/build_dataset.py`	Batch-processes all 35 (subject, session) pairs and writes one Parquet file per session plus a combined `full_dataset.parquet`. *Deliverable of Person 1.*
`src/capture/webcam.py`	Placeholder for the webcam drowsiness detector (MediaPipe FaceMesh, EAR, PERCLOS, yawn). *Owned by Person 3.*
`src/capture/watch.py`	Placeholder for reading smartwatch data via BLE. *Owned by Person 2.*
`src/features/watch_features.py`	Placeholder for the 60-second sliding-window feature extractor (HR mean, RMSSD, temperature trend, EDA peaks, motion variance). *Owned by Person 2.*
`src/features/video_features.py`	Placeholder for eye and mouth feature extraction. *Owned by Person 3.*
`src/features/fusion.py`	Placeholder for late fusion of watch and video probabilities. *Owned by Person 3.*
`src/models/baseline.py`	Placeholder for the gradient-boosting baseline with leave-one-subject-out evaluation. *Owned by Person 2.*
`src/models/gru.py`	Placeholder for the temporal GRU over the last 10 minutes. *Owned by Person 3.*
`src/models/calibration.py`	Placeholder for isotonic regression calibration. *Owned by Person 2.*
`src/advisor/nemotron_client.py`	Placeholder for the Nemotron 3.5 Lightning advisor running locally on the GX10 via vLLM. *Owned by Person 3.*
`src/advisor/fallback.py`	Rule-based fallback if the LLM output fails schema validation. *Owned by Person 3.*
`src/dashboard/app.py`	Placeholder for the Streamlit jury dashboard (risk timeline, threshold slider, trade-off curve). *Owned by Person 4.*
`src/driver_app/app.py`	Placeholder for the full-screen driver app (traffic light, spoken alert, "I'm alert" button). *Owned by Person 4.*

Path	Purpose
tests/	Unit and integration tests (currently empty).

6.3 What Is Deliberately Excluded

The following are **not** on GitHub:

Path	Reason
data/ (18 GB)	Raw UL-DD signals, aligned Parquet files, and extracted features. Too large for GitHub; distributed separately or downloaded by each teammate.
reference/ (95 MB)	Third-party reference notebooks from other teams. Excluded to keep the repository focused on our own code.
.venv/	Local Python virtual environment, machine-specific.
.DS_Store	macOS metadata files.
*.csv, *.parquet, *.h5, *.pt, *.mp4, *.zip	Large data and model artefacts, matched by pattern.
.env, *.pem, *.key	Secrets and credentials, never committed.

6.4 How to Reproduce the Environment

```
git clone git@github.com:fleurinejaegler/DrozyAI.git
cd DrozyAI
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

To reproduce the aligned dataset, each teammate must obtain the UL-DD raw files from Zenodo and place them under data/raw/CSV_Files/CSV_Files/, then run:

```
python -m src.main
```

This regenerates data/processed/full_dataset.parquet (84,000 rows) locally, without any need for GitHub to store the data.

6.5 Commit History

Commit	Message
a5d88c3	Fix .gitignore anchors and add Person 1 pipeline code
0a240a7	Implement Person 1 pipeline: align raw UL-DD to per-second table
7d32086	Set up project skeleton for DrozyAI drowsiness detection

Earlier history includes the repository initialisation and cleanup of the empty folders and stray .git directory.

6.6 Contributing Workflow

Each teammate works on their own machine and pushes to a feature branch, then opens a pull request:

```
git checkout -b person2/wearable-baseline
# ... edit code ...
git add src/models/baseline.py
git commit -m "Implement gradient boosting baseline"
git push -u origin person2/wearable-baseline
```

The team lead reviews the pull request and merges to `main`. Direct pushes to `main` are used only for small fixes by the data owner.